

MIT 2.003 SC PS6 Solving
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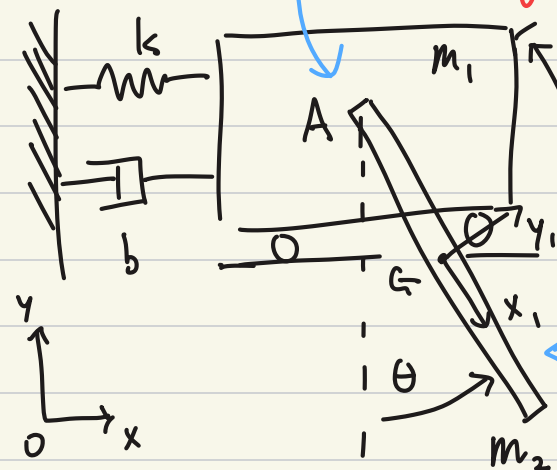
Given:

$$I = \int r^2 dm$$

$$\begin{cases} T \text{ is KE} \\ U \text{ is PE} \end{cases}$$

$$T_{(KE)} = \frac{1}{2} m \vec{v}_{G/O} \cdot \vec{v}_{G/O} + \frac{1}{2} \vec{\omega} \cdot \vec{I}_{G/O}$$

Problem 1: *cart*
crux is finding T_{rod} $\downarrow g$



Steps:

- 1: Find T_{cart}
- 2: Find T_{rod} *← harder*
- 3: Find V_{cart} & V_{rod}

T_{cart} :

No rotation. Velocity constrained in one direction

Hence

$$T_{\text{cart}} = \frac{1}{2} m_1 \dot{x}^2$$

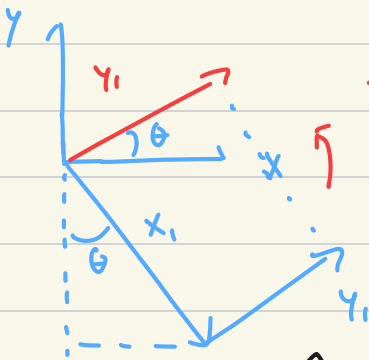
$$T_{rod} = \frac{1}{2} m_2 \vec{v}_{G/O} \cdot \vec{v}_{G/O} + \frac{1}{2} \vec{\omega} \cdot \vec{H}_{G/O}$$

L is pos
vector
 $\omega = \dot{\theta}$
because only
rotation about
a axis.

$$\begin{aligned} \vec{v}_{G/O} &= \vec{v}_{A/O} + \vec{v}_{G/A} \\ &= \dot{x} \hat{i} + \left(\frac{dL_{GA}}{dt} \right)_{\omega=0} + \omega \times L_{G/A} \\ &= \dot{x} \hat{i} + 0 + \dot{\theta} \hat{k} \times \frac{L}{2} \hat{i} \\ &= \dot{x} \hat{i} + \dot{\theta} \frac{L}{2} \hat{j} \end{aligned}$$

intermediate step:

express $\hat{j}_1$ in terms of $\hat{i}$ and $\hat{j}$



$$\therefore \hat{j}_1 = \cos \theta \hat{i} + \sin \theta \hat{j}$$

$$= \dot{x} \hat{i} + \dot{\theta} \frac{L}{2} \hat{j}_1$$

$$= \dot{x} \hat{i} + \dot{\theta} \frac{L}{2} (\cos \theta \hat{i} + \sin \theta \hat{j})$$

$$\vec{v}_{G/O} = \left(\dot{x} + \dot{\theta} \frac{L}{2} \cos \theta \right) \hat{i} + \dot{\theta} \frac{L}{2} \sin \theta \hat{j}$$

$$\vec{v}_{G/O} \cdot \vec{v}_{G/O} = \left(\dot{x} + \dot{\theta} \frac{L}{2} \cos \theta \right)^2 + \dot{\theta}^2 \frac{L^2}{4} \sin^2 \theta$$

$$= \dot{x}^2 + \dot{\theta} \dot{x} L \cos \theta + \dot{\theta}^2 \frac{L^2}{4} \cos^2 \theta + \dot{\theta}^2 \frac{L^2}{4} \sin^2 \theta$$

$$\therefore \cos^2 \theta + \sin^2 \theta \equiv 1$$

$$\vec{v}_{G/O} \cdot \vec{v}_{G/O} = \dot{x}^2 + \dot{x}L\dot{\theta}\cos\theta + \frac{\dot{\theta}^2 L^2}{4}$$

$$\frac{1}{2} m \vec{v}_{G/O} \cdot \vec{v}_{G/O} = \frac{1}{2} m (\dot{x}^2 + \dot{x}L\dot{\theta}\cos\theta + \frac{\dot{\theta}^2 L^2}{4})$$

For the second term: $\frac{1}{2} \dot{\theta} \hat{k} \cdot \vec{H}_{/G}$

$$\vec{H}_{/G} = [I] \{w\}$$

$$\{w\} = \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta} \hat{k} \end{Bmatrix}$$

Although

$$[I] = \begin{bmatrix} I_{xx/G} & \dots & \dots \\ \vdots & \ddots & \vdots \\ \vdots & \vdots & I_{zz/G} \end{bmatrix}$$

Because $\{w\}$ only has a $\hat{k}$ term. Everything except $I_{zz/G}$ will be never used.

Hence

$$\vec{H}_{/G} = I_{zz/G} \dot{\theta} \hat{k}$$

Intermediate step. Finding $I_{zz/G}$

$$I_{zz/G} = \int (x^2 + y^2) dm, \quad \because \text{slender rod. } y=0 \text{ always.}$$

$$I_{zz/G} = \int x^2 dm.$$

$$\text{density} = \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 dm.$$

$$dm = \sigma dA$$

$$= \frac{M}{L} dx$$

$$\rightarrow \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 dx$$

$$= \frac{M}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 dx$$

$$= \frac{M}{L} \frac{1}{3} \left[x^3 \right]_{-\frac{L}{2}}^{\frac{L}{2}}$$

$$= \frac{M}{3L} \left[\frac{L^3}{8} + \frac{L^3}{8} \right]$$

$$= \frac{1}{12} ML^2$$

$$I_{zz/G} = \frac{1}{12} ML^2$$

Rotational KE

$$= \frac{1}{2} \vec{\omega} \cdot \vec{H}_{I_G}$$

$$= \frac{1}{2} \dot{\theta} \hat{k} \cdot \left(\frac{1}{12} M L^2 \dot{\theta} \hat{k} \right)$$

$$= \frac{1}{24} M L^2 \dot{\theta}^2$$

$$T_{\text{rod}} = \frac{1}{2} m_2 \left[\dot{x}^2 + \frac{\dot{\theta}^2 L^2}{4} + \dot{x} \dot{\theta} L \cos \theta \right] + \frac{1}{24} M L^2 \dot{\theta}^2$$

$$= \frac{1}{2} m_2 \left[\dot{x}^2 + \frac{\dot{\theta}^2 L^2}{3} + \dot{x} \dot{\theta} L \cos \theta \right]$$

$$T_{\text{total}} = \frac{1}{2} (m_1 + m_2) \dot{x}^2 + \frac{1}{2} m_2 \left[\frac{\dot{\theta}^2 L^2}{3} + \dot{x} \dot{\theta} L \cos \theta \right]$$

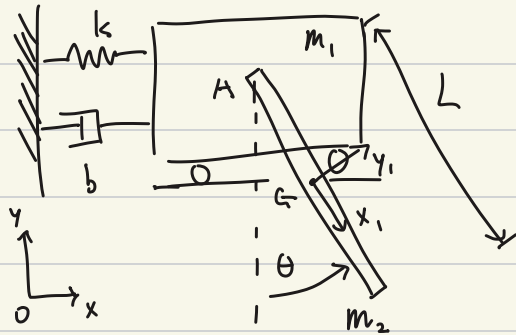
PE time:

V_{cart}

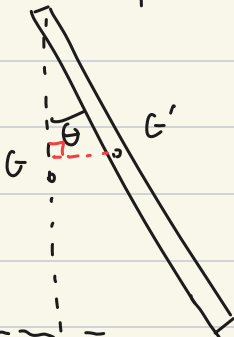
Really just from the Spring.

$$V_{\text{cart}} = \frac{1}{2} k x^2$$

where $x=0$ is equilibrium position.



V_{rod} : Only from $mg \Delta h$ that G experiences.



$$G \text{ height} = \frac{L}{2}$$

$$G' \text{ height} = \frac{L}{2} \cos \theta$$

$$\Delta h = \frac{L}{2} - \frac{L}{2} \cos \theta$$

$$= \frac{L}{2} (1 - \cos \theta)$$

$$V_{\text{rod}} = m_2 g \frac{L}{2} (1 - \cos \theta)$$

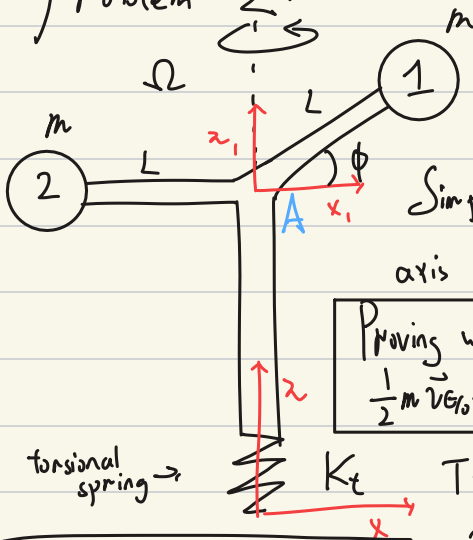
set $y=0$

Ans problem 1:

$$V_{\text{total}} = \frac{1}{2} k x^2 + m_2 g \frac{L}{2} (1 - \cos \theta)$$

$$T_{total} = \frac{1}{2}(m_2 + m_1)\dot{x}^2 + \frac{1}{2}m_2 \left[\frac{\dot{\theta}^2 L^2}{3} + L\dot{x}\dot{\theta}\cos\theta \right]$$

Problem 2:



Steps:

1. Find T

2. Find V through K_t .

Simplification rule: If body rotates about a fixed axis through pt A. $T = \frac{1}{2} \vec{\omega} \cdot \vec{H}_A$

Proving why

$$\frac{1}{2} m \vec{v}_{G/O} \cdot \vec{v}_{G/O} + \frac{1}{2} \vec{\omega} \cdot \vec{H}_A = \frac{1}{2} \vec{\omega} \cdot \vec{H}_A$$

torsional spring $\rightarrow$

$$T = \sum \frac{1}{2} m_i (v_i \cdot v_i)$$

v_i is relative to an inertial frame.

$$v_i = \omega \times r_i / A$$

$$T = \frac{1}{2} \sum m_i v_i \cdot (\omega \times r_{i/A})$$

$$\therefore a \cdot (b \times c) = b \cdot (c \times a)$$

$$T = \frac{1}{2} \sum m_i \omega \cdot (r_{i/A} \times v_i)$$

$$= \frac{1}{2} \omega \cdot \sum (r_{i/A} \times v_i)$$

$$= \frac{1}{2} \vec{\omega} \cdot \vec{H}_A \quad ||$$

Finding T

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{H}|_A$$

$$\frac{1}{H|_A} = \sum_{j=1}^n (-\hat{v}_j) \times m L(R(\hat{v}_j)) + \dots = m L^2 \Omega_k + \dots$$

Obj 2 term in $\vec{H}_A$

$$\Omega_{2/A} \times P_{2/0}$$

$$\begin{aligned} \vec{v}_{2/A} &= \left(\frac{d\vec{r}_{2/A}}{dt} \right)_{w=0} + \vec{\omega} \times \vec{r}_{2/A} \\ &= 0 + \Omega \hat{k} \times (L \cos \theta \hat{i}_1 + L \sin \theta \hat{k}) \\ &= \Omega L \cos \theta \hat{j}_1 \end{aligned}$$

$$\begin{aligned} \Omega_{2/A} \times P_{2/0} &= (L \cos \theta \hat{i}_1 + L \sin \theta \hat{k}) \times m \Omega L \cos \theta \hat{j}_1 \\ &= m \Omega L^2 \cos^2 \theta \hat{k} + 0 \\ &= m \Omega L^2 \cos^2 \theta \hat{k} \end{aligned}$$

$$\begin{aligned} H_A &= m \Omega L^2 \hat{k} + m \Omega L^2 \cos^2 \theta \hat{k} \\ &= m \Omega L^2 (1 + \cos^2 \theta) \hat{k} \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \vec{\omega} \cdot H_A &= \frac{1}{2} \Omega \hat{k} \cdot m \Omega L^2 (1 + \cos^2 \theta) \hat{k} \\ &= \frac{1}{2} m \Omega^2 L^2 (1 + \cos^2 \theta) \end{aligned}$$

$$T = \frac{1}{2} m \Omega^2 L^2 (1 + \cos^2 \theta)$$

 Finding V

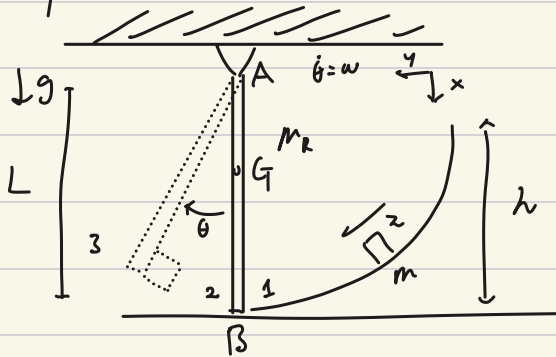
Only component: Torsional spring where $\tau = K_t \theta$
Hence $V = \frac{1}{2} K_t \theta^2$ where $\theta = 0$ is

equilibrium position.

 Ans

$$\begin{aligned} V &= \frac{1}{2} K_t \theta^2 \\ T &= \frac{1}{2} m \Omega^2 L^2 (1 + \cos^2 \theta) \end{aligned}$$

Problem 3:



Steps

- 1: Find GPE lost to KE at position 1
- 2: Find KE at position 2 (cross)
- 3: Find GPE at position 3, you could use conservation of E or just find it.

1: KE of Z at position 1

$$T_0 + V_0 = T_1 + V_1$$

$$0 + V_0 = T_1 + 0$$

$$V_0 = T_1$$

$$T_1 = mgh$$

2: Rotational Kinetic Energy at position 2.

$$\text{Use } \frac{1}{2} \vec{\omega} \cdot \vec{H}_{I/A} \text{ (proved in problem 2, p6)}$$

$$\vec{H}_{I/A} = I_{zz/A} \cdot \vec{\omega} \text{ (proved in problem 1, p4)}$$

$$I_{\text{total}} = I_{\text{no z}} + I_{\text{no rod}}$$

Finding $I_{\text{no z}}$

Proving parallel axis theorem

$$\int r^2 dm \text{ where } r \text{ is st line dist from COR to mass location}$$

$$= \int (x^2 + y^2) dm$$

e.g. $\int_{x,y} \dots$ fixed to rotate at A

$$I_A = \int ((x+d)^2 + y^2) dm$$

$$\begin{aligned}
 I_{/A} &= \int ((x+d)^2 + y^2) dm \\
 &= \int (x^2 + 2xd + d^2 + y^2) dm \\
 &= \int (x^2 + y^2) dm + 2d \int x dm + \int d^2 dm
 \end{aligned}$$

Since d is a constant, $\int x dm = 0$
as $x=0$ was defined at COM.

$$= \int (x^2 + y^2) dm + 0 + md^2$$

$$I_{/A} = I_{cm} + md^2$$

$I_{no\ z}$ (Assuming massless rod with no z attached to end)

$$\begin{aligned}
 I_{/cm}^{no\ z} &= \int r^2 dm \\
 &= \int_{-\frac{L}{2}}^{\frac{L}{2}} r^2 dm \\
 &= \frac{M_R}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} r^2 dr \\
 &= \frac{M_R}{3L} \left[r^3 \right]_{-\frac{L}{2}}^{\frac{L}{2}} \\
 &= \frac{1}{12} M_R L^2
 \end{aligned}$$

$$\begin{aligned}
 dm &= \sigma dr \\
 &= \frac{M_R}{L} dr
 \end{aligned}$$

$$\begin{aligned}
 I_{/A}^{no\ rod} &= \int L^2 dm \\
 &= mL^2 \quad (L \text{ is constant})
 \end{aligned}$$

$$I_{total\ /A} = I_{/A}^{no\ z} + I_{/A}^{no\ rod}$$

$$\begin{aligned}
 &= \frac{1}{3} M_R L^2 + mL^2 \\
 &= L^2 \left(\frac{1}{3} M_R + m \right)
 \end{aligned}$$

Parallel axis theorem

$$\begin{aligned}
 I_{/A}^{no\ z} &= \frac{1}{12} M_R L^2 + M_R \left(\frac{L}{2} \right)^2 \\
 &= \frac{1}{12} M_R L^2 + \frac{1}{4} M_R L^2 \\
 &= \frac{1}{3} M_R L^2
 \end{aligned}$$

$$I_{zz/A} = L^2 \left(\frac{1}{3} M_R + m \right)$$

$$\vec{H}_{/A} = L^2 \left(\frac{1}{3} M_R + m \right) \omega \hat{k}$$

Hence rotational KE: $\frac{1}{2} \vec{H}_{/A} \cdot \vec{\omega}$

$$= \frac{1}{2} L^2 \left(\frac{1}{3} M_R + m \right) \omega^2 \quad \text{At position 2, 0 PE}$$

At position 3, some time after collision.

$$T = \frac{1}{2} L^2 \left(\frac{1}{3} M_R + m \right) \omega^2$$

$$V = PE_{\text{mass 2}} + PE_{\text{rod}}$$

$$= m g (L - L \cos \theta) + M_R g \left(\frac{1}{2} - \frac{L}{2} \cos \theta \right)$$

$$= (L - L \cos \theta) \left(m g + M_R g \frac{1}{2} \right)$$

$$= L g (1 - \cos \theta) \left(m + M_R \frac{1}{2} \right)$$

! you can use

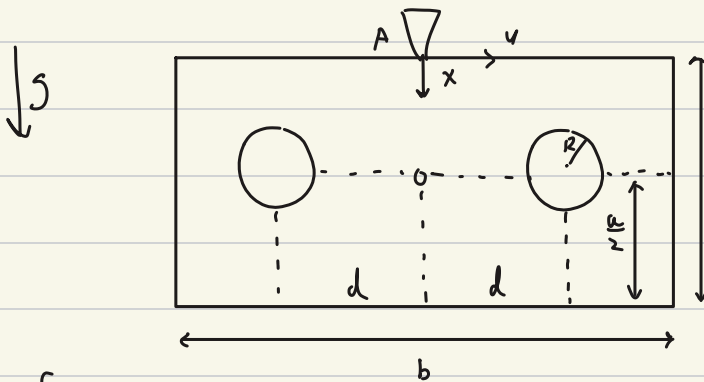
conservation of E to

find values, for e.g.

finding max $\angle$ with

$$T_2 = V_{\text{max}}$$

Problem 4: thickness = t density = ρ



a) Find T, V

b) Find # dof

(degrees of freedom)

Steps

1: Find $I_{/A}$ (the crux)

2: Find T

3: Find V

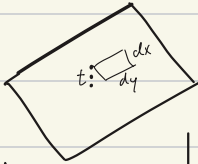
Finding I_A (the crux)

Decompose $I_A = \overset{\text{no holes}}{I_A} - \overset{\text{only filled holes}}{I_A}$

Decompose $\overset{\text{no holes}}{I_A} = \overset{\text{no holes}}{I_{CM}} + \overset{\text{no holes}}{M} \left(\frac{a}{2}\right)^2$ Parallel axis thm. Problem 3 p.9

$\overset{\text{no holes}}{I_{CM}} = \int (x^2 + y^2) dm$

$dm = \rho t dx dy$



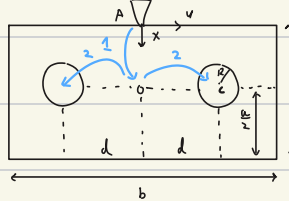
$$\begin{aligned} \overset{\text{no holes}}{I_{CM}} &= \rho t \int_{-\frac{a}{2}}^{\frac{a}{2}} \int_{-\frac{b}{2}}^{\frac{b}{2}} (x^2 + y^2) dx dy \\ &= \rho t \int_{-\frac{a}{2}}^{\frac{a}{2}} \left[\frac{x^3}{3} + y^2 x \right]_{-\frac{b}{2}}^{\frac{b}{2}} dy \\ &= \rho t \int_{-\frac{a}{2}}^{\frac{a}{2}} \left(\frac{1}{12} a^3 + y^2 a \right) dy \\ &= \rho t \left[\frac{1}{12} a^3 y + \frac{y^3 a}{3} \right]_{-\frac{a}{2}}^{\frac{a}{2}} \\ &= \rho t \left[\frac{1}{12} a^3 b + \frac{1}{12} b^3 a \right] \\ &= \frac{1}{12} a b \rho t (a^2 + b^2) \end{aligned}$$

$$\begin{aligned} \overset{\text{no holes}}{I_A} &= \frac{1}{12} a b \rho t (a^2 + b^2) + a b t \rho \frac{a^2}{4} \\ &= a b \rho t \left(\frac{1}{3} a^2 + \frac{1}{12} b^2 \right) \\ &= \frac{1}{3} a b \rho t \left(a^2 + \frac{1}{4} b^2 \right) \end{aligned}$$

Parallel axis thm. Problem 3 p.9

Decompose $\overset{\text{only filled holes}}{I_A} = \overset{\text{only filled holes}}{I_{CM}} + \overset{\text{only filled holes}}{M_{\text{holes}}} \left(\frac{a}{2}\right)^2$

Decompose $\overset{\text{only filled holes}}{I_{CM}} = 2 \left(\overset{\text{one hole, filled}}{I_{CM_C}} + M_{\text{one hole}} d^2 \right)$



$M_{\text{one hole}} d^2 = \rho \pi R^2 d^2$

$\overset{1 \text{ hole, filled}}{I_{CM_C}} = \int_0^R r^2 dm \quad dm = t \rho 2\pi r dr$

$= 2\pi \rho t \int_0^R r^3 dr$

$= \frac{t\pi \rho}{4} R^4$

$\overset{2 \text{ holes, filled}}{I_{CM}} = 2 \left(\frac{t\pi \rho}{4} R^4 + t \rho \pi R^2 d^2 \right)$

$= t 2 \rho \pi R^2 \left(R^2 \frac{1}{4} + d^2 \right)$

$= t \rho \pi R^2 \left(\frac{1}{2} R^2 + d^2 \right)$

$\overset{\text{only filled holes}}{I_A} = 2 t \rho \pi R^2 \left(\frac{1}{2} R^2 + d^2 \right) + 2 \rho \pi R^2 t \left(\frac{a^2}{2} \right)$

$= 2 \rho \pi R^2 t \left(\frac{1}{2} R^2 + d^2 + \frac{a^2}{2} \right)$

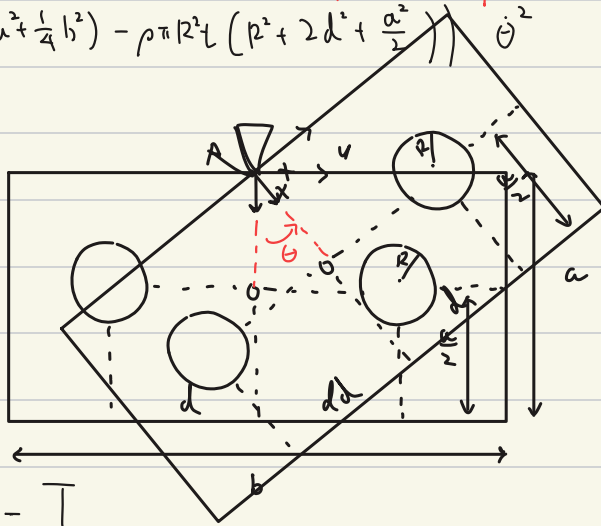
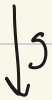
$$I_{/A} = \frac{1}{3} ab \rho t (a^2 + \frac{1}{4} b^2) - \rho \pi R^2 t (R^2 + 2d^2 + \frac{a^2}{2})$$

$$T = \frac{1}{2} \vec{\omega} \cdot [I_{/A}] \{ \omega \}$$

$$= \frac{1}{2} I_{zz/A} \omega_z^2 \quad \text{where } \omega_z = \dot{\theta} \quad (\text{proved in problem 2 p. 6})$$

$$= \frac{1}{2} \left(\frac{1}{3} ab \rho t (a^2 + \frac{1}{4} b^2) - \rho \pi R^2 t (R^2 + 2d^2 + \frac{a^2}{2}) \right) \dot{\theta}^2$$

Finding V



Decompose $T = T_{\text{full block}} - T_{\text{2 circles.}}$

Answer Problem 4 a)

$$T_{\text{full block}} = M g \left(\frac{a}{2} - \frac{a}{2} \cos \theta \right)$$

$$= M g \frac{a}{2} (1 - \cos \theta)$$

$$T_{\text{2 circles}} = 2 m g \left(\frac{a}{2} - \frac{a}{2} \cos \theta \right)$$

$$= 2 m g \frac{a}{2} (1 - \cos \theta)$$

$$T = (M - 2m) g \frac{a}{2} (1 - \cos \theta)$$

$$= \rho t (ab - 2\pi R^2) g \frac{a}{2} (1 - \cos \theta)$$

$$T = \frac{1}{2} \left(\frac{1}{3} ab \rho t (a^2 + \frac{1}{4} b^2) - \rho \pi R^2 t (R^2 + 2d^2 + \frac{a^2}{2}) \right) \dot{\theta}^2$$

$$V = \rho t (ab - 2\pi R^2) g \frac{a}{2} (1 - \cos \theta)$$

Problem 4 b) :

$$\# \text{ dof} = 6(\# \text{ rigid bodies}) + 3(\# \text{ pt}) - \# \text{ constraints}$$

$$N=0, \quad N=1 \quad C=?$$

problem 4 b): ANS

Finding C:

No x, y axis rotation. 2

No x, y, z axis translation 3

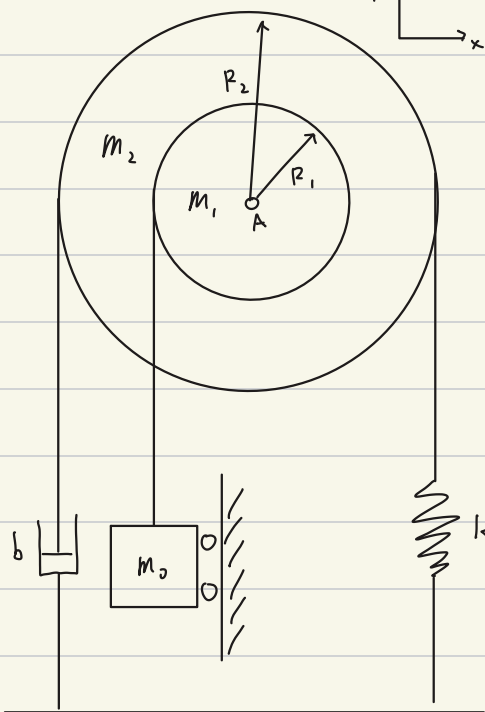
$$\therefore 6 - 3 - 2 = 1$$

$$\# \text{ dof} = 1$$

Constraints: $\dot{x}, \dot{y}, \dot{z}$, x & y axis rotation all = 0

Problem 5:

find T & V



Step 1: Find V (crux)

Step 2: Find T

Assume θ_{static} where $\sum \tau = 0$

$$\sum \tau = m_0 g R_1 \hat{k} - k R_2^2 \theta_{\text{static}} \hat{k} = 0$$

$$\theta_{\text{static}} = \frac{m_0 g R_1}{k R_2}$$

Assume if $\theta_0 = 0$, equilibrium is achieved.

$$V = \sum \text{-work done by conservative ext } F\text{'s (cart, springs etc)}$$

$$\begin{aligned} &= \int_0^{\theta_d} (-m_0 g R_1 + k R_2^2 (\theta_d + \theta_{\text{static}})) d\theta_d \\ &= \int_0^{\theta_d} (-m_0 g R_1 + k R_2^2 \theta_{\text{static}} + k R_2^2 \theta_d) d\theta_d \\ &= \int_0^{\theta_d} (0 + k R_2^2 \theta_d) d\theta_d \\ &= \frac{1}{2} k R_2^2 [\theta_d^2]_0^{\theta_d} \\ &= \frac{1}{2} k R_2^2 \theta_d^2 \end{aligned}$$

Say $\dot{\theta} = \omega$

$$V = \frac{1}{2} k R_2^2 \theta_d^2 \quad \text{where } \theta_d = 0 \text{ is equilibrium.}$$

$$T = T_{\text{cart}} + T_{\odot}$$

$$T_{\text{cart}} = \frac{1}{2} m_0 \dot{y}^2$$

$$T_{\odot} = \frac{1}{2} \vec{\omega} \cdot \vec{H}_{/A} \quad (\text{proved in problem 2 p. 6})$$

$$\vec{H}_{/A} = \vec{I}_{/A} \vec{\omega}$$

$$\vec{I}_{/A} = \vec{I}_{/\text{small disc A}} + \vec{I}_{/\text{big disc A}}$$

$$\begin{aligned} I_{/\text{small disc A}} &= \int_{R_1} r^2 dm \\ &= \int_0^{R_1} r^2 dm \\ &= \frac{2m_1}{R_1^2} \int_0^{R_1} r^3 dr \\ &= \frac{2m_1}{R_1^2} \left[\frac{r^4}{4} \right]_0^{R_1} \\ &= \frac{1}{2} m_1 R_1^2 \end{aligned}$$

$$dm = \rho 2\pi r dr$$

$$= \frac{2m_1}{R_1^2} r dr$$

$$\begin{aligned} I_{/\text{big disc A}} &= \int_0^{R_2} r^2 dm \\ &\vdots \quad \text{same} \\ &= \frac{1}{2} m_2 R_2^2 \end{aligned}$$

$$\begin{aligned} I_{/A} &= \frac{1}{2} m_1 R_1^2 + \frac{1}{2} m_2 R_2^2 \\ &= \frac{1}{2} (m_1 R_1^2 + m_2 R_2^2) \end{aligned}$$

$$\vec{H}_{/A} = \frac{1}{2} (m_1 R_1^2 + m_2 R_2^2) \vec{\omega} \uparrow k$$

$$T_{\odot} = \frac{1}{4} (m_1 R_1^2 + m_2 R_2^2) \omega^2$$

$$T = \frac{1}{2} m_0 \dot{y}^2 + \frac{1}{4} (m_1 R_1^2 + m_2 R_2^2) \omega^2$$

Problem 5 solution:

$$T = \frac{1}{2} m_o \dot{\gamma}^2 + \frac{1}{4} (m_1 R_1^2 + m_2 R_2^2) \omega^2$$

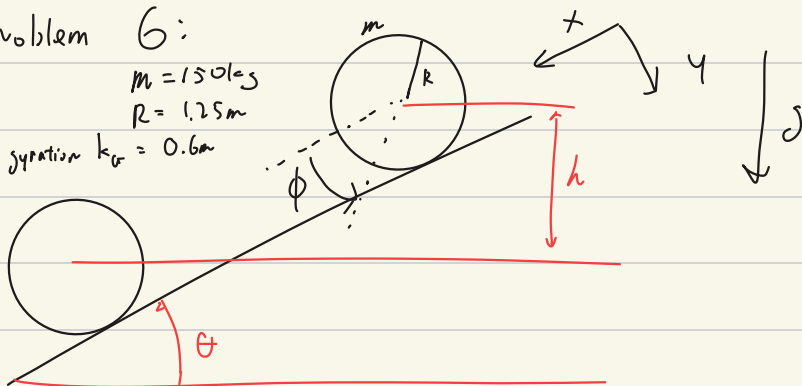
$$V = \frac{1}{2} k R_2^2 \theta_d^2$$

Problem 6:

$$m = 150 \text{ kg}$$

$$R = 1.25 \text{ m}$$

$$R \text{ of gyration } k_G = 0.6 \text{ m}$$



a) After h descended, find T .

$\therefore$ GPE loss = KE gained.

$$KE_{\text{translational}} + KE_{\text{rotational}} = \text{GPE}$$

Step 1: $KE_{\text{translational}}$

Step 2: $KE_{\text{rotational}}$

b) Ratio of $KE_{\text{rot}} : KE_{\text{total}}$

Step 1: just solve it by

cancelling terms.

Finding $KE_{\text{translational}}$ or KE_{rot} for short

$\therefore$ No slip condition.

$$d = R \phi$$

$$v = R \dot{\phi}$$

$$KE_{\text{rot}} = \frac{1}{2} m v^2$$

$$= \frac{1}{2} m (R \dot{\phi})^2$$

Finding $KE_{\text{rotational}}$, KE_{rot} .

$$KE_{\text{rot}} = \frac{1}{2} \vec{\omega} \cdot \vec{H}_A \quad (\text{proved in problem 2, p.6})$$

$$\vec{\omega} = \begin{Bmatrix} 0 \\ 0 \\ \dot{\phi} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \dot{\phi} \end{Bmatrix}$$

$$\vec{H}_A = I_A \cdot \vec{\omega}$$

$$= m k_G^2 \dot{\phi}$$

$$\text{Definition: } I = m k_G^2$$

where k_G = radius of gyration.

$$KE_p = \frac{1}{2} k_G^2 m \dot{\phi}^2$$

$$KE = \frac{1}{2} m R^2 \dot{\phi}^2 + \frac{1}{2} m k_G^2 \dot{\phi}^2$$

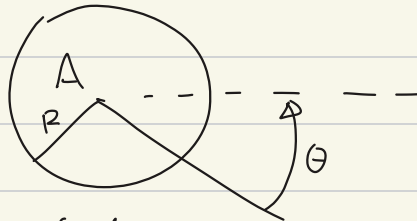
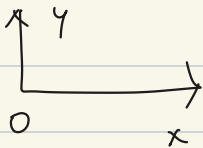
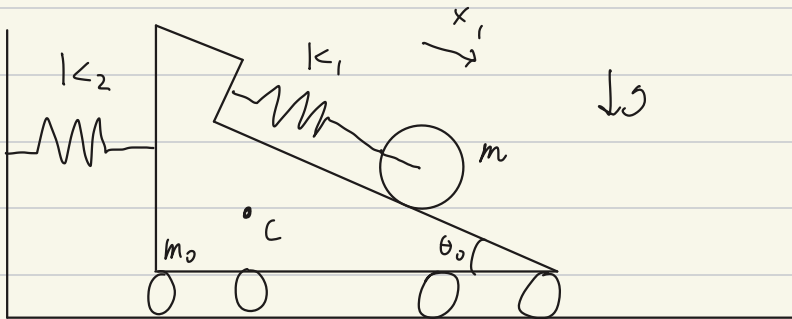
Answer:

$$\frac{KE_{\text{total}}}{|K_{\text{total}}|} = \frac{\frac{1}{2} m (R \dot{\phi})^2}{\frac{1}{2} m (R \dot{\phi})^2 + \frac{1}{2} m (k_G \dot{\phi})^2}$$

$$= \frac{R^2}{R^2 + k_G^2}$$

Problem 7

mass = m_0



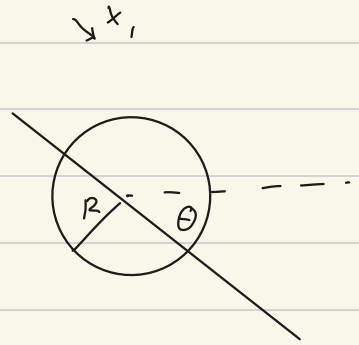
a) Find T, V

b) Use Lagrange equations to find EOM of system

Finding T & v

Step 1: find T .

There are two rigid bodies with their kinetic energy to consider. The cart only has translational KE, while the disk has both translational and rotational.



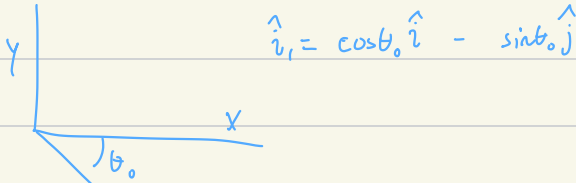
Cart translational KE_T :

$$\frac{1}{2} m_0 \dot{x}^2$$

Disk KE:

$$\frac{1}{2} m \vec{v}_{A/O} \cdot \vec{v}_{A/O} + \frac{1}{2} \vec{\omega} \cdot \vec{H}_{A/O}$$

$$\begin{aligned} \vec{v}_{A/O} &= \vec{v}_{A/C} + \vec{v}_{C/O} \\ &= \dot{x}_1 \hat{i} + \dot{x} \hat{i} \end{aligned}$$



$$\begin{aligned} &= \frac{1}{2} m \left(\dot{x}_1 (\cos \theta_0 \hat{i} - \sin \theta_0 \hat{j}) + \dot{x} \hat{i} \right) \cdot \left(\dot{x}_1 (\cos \theta_0 \hat{i} - \sin \theta_0 \hat{j}) + \dot{x} \hat{i} \right) + \dots + \frac{1}{2} m_0 \dot{x}^2 \\ &= \frac{1}{2} m \left(\dot{x}_1^2 \cos^2 \theta_0 + 2 \dot{x} \dot{x}_1 \cos \theta_0 + \dot{x}^2 + \dot{x}_1^2 \sin^2 \theta_0 \right) + \dots + \frac{1}{2} m_0 \dot{x}^2 \\ &= \frac{1}{2} m \left(\dot{x}_1^2 + 2 \dot{x} \dot{x}_1 \cos \theta_0 + \dot{x}^2 \right) + \dots + \frac{1}{2} m_0 \dot{x}^2 \end{aligned}$$

$$\frac{1}{2} \vec{\omega} \cdot \vec{H}_{A/O} \quad (A \text{ is com})$$

$$\vec{H}_{A/O} = \vec{I}_{A/O} \cdot \vec{\omega}$$

$$\vec{\omega} = \begin{Bmatrix} 0 \\ 0 \\ \omega \end{Bmatrix}$$

$$I_{/A} = \int r^2 dm (\text{disk})$$

$$dm = \rho 2\pi r dr$$

$$= \frac{m}{\pi R^2} 2\pi r dr$$

$$= \frac{2m}{R^2} r dr$$

$$I_{/A} = \frac{2m}{R^2} \int_0^R r^3 dr$$

$$= \frac{2m}{2R^2} [R^4]$$

$$= \frac{1}{2} m R^2$$

$$\vec{H}_{/A} = \frac{1}{2} m R^2 \vec{\omega}$$

$$\frac{1}{2} \vec{\omega} \cdot \vec{H}_{/A} = \frac{1}{4} m R^2 \omega^2$$

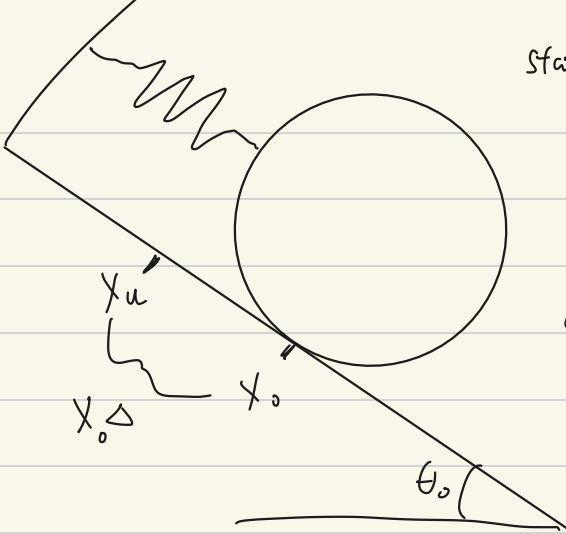
$$\begin{aligned} T &= \frac{1}{4} m R^2 \omega^2 + \frac{1}{2} m (\dot{x}_1^2 + 2\dot{x}_1 \dot{x}_2 \cos \theta_0 + \dot{x}_2^2) + \frac{1}{2} m_0 \dot{x}^2 \\ &= \frac{1}{4} m \dot{x}_1^2 + \frac{1}{2} m (\dot{x}_1^2 + 2\dot{x}_1 \dot{x}_2 \cos \theta_0) + \frac{1}{2} \dot{x}^2 (m_0 + m) \\ &= \frac{1}{2} m \left(\frac{3}{2} \dot{x}_1^2 + 2\dot{x}_1 \dot{x}_2 \cos \theta_0 \right) + \frac{1}{2} \dot{x}^2 (m_0 + m) \end{aligned}$$

Step 2: Finding V

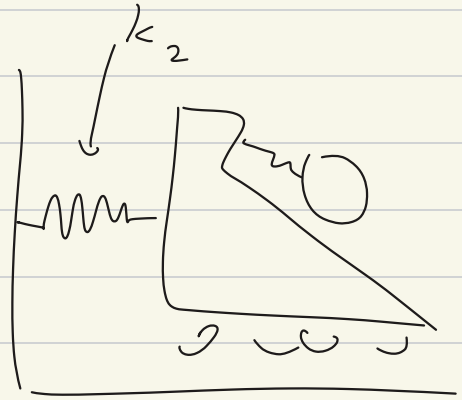
Let x_u be the displacement from the wall of unstretched length of the spring, call the static equilibrium position as x_0 .

(will work out both the static equilibrium (GPE & spring energy for better visualization of cancellation)

Static equil.



and



Ball

Cart

Let GPE at x_u be 0.
Hence GPE at static equilibrium
is $-mg \sin \theta_0 x_0$.
Elastic PE at equilibrium $+ \Delta x$
 $\frac{1}{2} k_1 (x_0 + \Delta x)^2$

Cart PE is just
 $\frac{1}{2} k_2 (\Delta x)^2$
where spring force at
 $\Delta x = 0$ is 0

Hence

$$V = -mg \sin \theta_0 x_0 + \frac{1}{2} k_1 (x_0 + \Delta x)^2 + \frac{1}{2} k_2 (\Delta x)^2$$

$$T = \frac{1}{2} m \left(\frac{3}{2} \dot{x}_1^2 + 2 \dot{x} \dot{x}_1 \cos \theta_0 \right) + \frac{1}{2} \dot{x}^2 (m_0 + m)$$

b) Finding EOMs with Lagrange

$$\mathcal{L} = \frac{1}{2}m \left(\frac{3}{2} \dot{x}_1^2 + 2\dot{x}_1 \dot{x} \cos\theta_0 \right) + \frac{1}{2} \dot{x}^2 (m_0 + m) - \left(-mg \sin\theta_0 x_0 \Delta + \frac{1}{2}k_1 (x_0 \Delta_1 + \Delta x_1)^2 + \frac{1}{2}k_2 (\Delta x)^2 \right)$$

where Δx is one in x_1 not x coordinates, except

As $\mathcal{L} = T - V$

Just 2 coordinates. x_1 and x the k_2 term

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{d}{dt} \left(\frac{1}{2}m 2\dot{x} \cos\theta_0 + \dot{x}(m_0 + m) - (0) \right) - k_2 x = 0$$

$$\text{1st EOM: } m\ddot{x}_1 \cos\theta_0 + \ddot{x}(m_0 + m) - k_2 x = 0$$

Now with x_1

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}_1} \right) - \frac{\partial \mathcal{L}}{\partial x_1} = 0$$

$$\frac{d}{dt} \left(\frac{1}{2}m(3\dot{x}_1 + 2\dot{x} \cos\theta_0) \right) - \left(-mg \sin\theta_0 + k_1(x_0 \Delta_1 + \Delta x_1) \right) = 0$$

Observe that $-mg \sin\theta_0 + k_1 x_0 \Delta_1 = 0$.

Since $x_0 \Delta_1$ is equilibrium. Downward force Force of spring

$$\text{1st EOM: } m\ddot{x}_1 \cos\theta_0 + \ddot{x}(m_0 + m) - k_2 x = 0$$

$$\text{2nd EOM: } \frac{3}{2}m\ddot{x}_1 + m\ddot{x} \cos\theta_0 - k_1 \Delta x_1 = 0$$

Thanks! Good luck to you :)